Contact

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Top Skills

LangChain
ChromaDB
Artificial Intelligence (AI)

Languages

English
Malayalam
Hindi

Certifications

Generative AI with Large Language Models
Claude Academy: Introduction to Model Context Protocol
AWS Certified AI Practitioner
Neural Networks and Deep Learning
Convolutional Neural Networks in TensorFlow

Publications

Predicting Suicidal Behavior Among Indian Adults Using Childhood Trauma, Mental Health Questionnaires and Machine Learning Cascade Ensembles

Classification of executive functioning performance post-longitudinal tDCS using functional connectivity and machine learning methods

Prediction of multitasking performance post-longitudinal tDCS via EEG-based functional connectivity and machine learning methods

Vishnu M

AI/ML Engineer | Generative AI & Agentic AI | LLM Evaluation, RAG & AI Reliability | Scalable ML Systems | AWS & MLOps
Buffalo, New York, United States

Summary

AI/ML Engineer with 4+ years of experience developing production machine learning and generative AI systems, specializing in AI reliability, LLM evaluation, and scalable inference infrastructure. Experienced in architecting multi-agent workflows, hallucination detection systems, self-verifying RAG pipelines, and enterprise model-serving platforms. Supported systems handling 1M+ daily requests with 99.9% uptime and improved inference performance through retrieval optimization, model quantization, and caching. Focused on building reliable, evaluation-driven AI systems through LLM fine-tuning, retrieval optimization, automated verification, and production monitoring.

Experience

C3 AI
AI/ML Engineer
August 2025 - Present (1 year 3 months)
United States

- Architected a multi-tenant LLM and RAG inference platform supporting 20+ enterprise deployments and 1M+ daily requests, separating retrieval, orchestration, and model-serving services on Kubernetes to maintain 99.9% uptime.
- Designed LangGraph-based AI orchestration workflows integrating document analysis, retrieval, LLM inference, and response verification, with automated validation gates, retry mechanisms, and fault isolation across enterprise deployments.
- Spearheaded the migration of 12 legacy ML workloads from on-premises infrastructure to AWS, containerizing services with Docker and managing cloud environments through Terraform; reduced infrastructure costs by 35% and deployment cycles from 10 days to 2 days.
- Orchestrated a cross-functional team of 6 engineers and 3 product owners to deliver a document-intelligence agent in 10 weeks, establishing evaluation criteria, milestone ownership, and phased production rollout procedures.

Prediction of decision-making performance post-longitudinal tDCS administration via EEG features and machine learning

Classification of Attention Performance Post-Longitudinal tDCS via Functional Connectivity and Machine Learning Methods

- Instituted automated model evaluation and drift-monitoring workflows using MLflow, Prometheus, and Grafana, integrating regression checks and monitoring alerts into CI/CD processes to detect model regressions 4x faster and reduce production incidents by 40%.
- Hardened embedding and vector-search infrastructure handling 50M+ documents through index sharding, response caching, and load-tested failover, maintaining p95 latency below 300 ms under 3x normal traffic.
- Mentored 4 junior engineers through weekly architecture reviews, production deployment walkthroughs, and code reviews, reducing their time to first production release from 8 weeks to 5.

Glean

AI/ML Engineer

January 2023 - July 2024 (1 year 7 months)

Bengaluru, Karnataka, India

- Optimized transformer-based semantic ranking models for enterprise search using hard-negative mining and embedding fine-tuning, improving offline top-3 relevance by 18% on NDCG and online click-through rates by 9%.
- Streamlined model development workflows through reusable Apache Airflow pipelines and standardized feature management, reducing training and evaluation turnaround from 5 days to 1.5 days.
- Investigated retrieval-quality regressions across 30+ enterprise data connectors, identifying index staleness and tokenization inconsistencies through query-level diagnostics and log analysis, restoring measured answer accuracy to 94%.
- Accelerated search inference performance through INT8 model quantization, request batching, and embedding caching, reducing p95 query latency by 40%, from 820 ms to 490 ms.
- Tuned RAG retrieval and generation configurations across a 1,500-question evaluation benchmark, experimenting with document chunk sizes, reranking, and citation-grounded prompting to reduce hallucinated responses by 27%.

Accenture

2 years

Data Scientist

August 2021 - December 2022 (1 year 5 months)

Kochi, Kerala, India

- Constructed a retail demand-forecasting model using XGBoost and Prophet ensembles, engineering 60+ predictive features and improving forecast accuracy by reducing MAPE from 18% to 14%.

- Integrated an NLP-based service ticket classifier with the client's operational platform using FastAPI and Docker on Azure, automating 70% of ticket routing and saving approximately 1,200 agent hours quarterly.
- Automated weekly operational reporting for 5 business units using Python, SQL, and Airflow pipelines, replacing manual Excel processes and saving 25 analyst hours per week.
- Formulated a randomized experimentation framework for customer-retention initiatives, defining cohort assignments and statistical significance tests that supported a 6% improvement in retention across 40K customers.

Associate Data Scientist

January 2021 - July 2021 (7 months)

Kochi, Kerala, India

- Prototyped a hybrid recommendation system combining collaborative and content-based filtering with Python, Pandas, and NumPy, achieving a 12% increase in click-through rates during A/B testing.
- Trained Random Forest and XGBoost churn prediction models, evaluating 40+ candidate features and performing cross-validated hyperparameter tuning to achieve an AUC of 0.86.
- Validated a BERT-based document-classification pilot on 10K labeled records, achieving a 91% F1 score through text preprocessing with SpaCy and NLTK and model version tracking in SageMaker.

Education

University at Buffalo

Master's degree, Computer Science · (August 2024 - January 2026)

APJ Abdul Kalam Technological University

Bachelor of Technology - BTech, Electrical, Electronics and Communications Engineering · (2017 - 2021)

SCMS School of Engineering & Technology

Bachelor of Technology - BTech, Electrical, Electronics and Communications Engineering · (2017 - 2021)